ROHAN SOGANI

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TECHNICAL EXPERTISE

Languages: Java, Go, Rust, Python, C/C++, SQL

Framework & Storage: Netty, FastAPI, Kafka, Dropwizard, Actix/Tokio, Oracle, MongoDB, Postgres, Redis/ValKey, Elastic Search

Cloud & Infra: OCI, AWS, GCP, Azure, Kubernetes, Istio, Helm, Docker, Terraform | **Tools:** Prometheus, Grafana, Locust, Git

PROFESSIONAL EXPERIENCE

Senior Member of Technical Staff | Oracle Cloud Infrastructure, Seattle, WA ***August 2022 – Present***

- **Architected** the control plane and data plane for OCI Generative AI Agent Service using **Kubernetes Operators** and **Knative + Istio**: eliminating noisy-neighbours, having **Redis-backed memory**, and **scale-to-zero** endpoints serving thousands of customers across **10,000+ agent endpoints** with **99.9% availability** and reduced idle infrastructure cost.
- **Led the migration** of OCI Generative AI's **OpenAI-compatible agent** execution from **Java Runtime to an independently deployable Rust/Tokio-based Shepherd Model Gateway** (**open source**), isolating streaming and agentic loop from data plane API tier, achieving 15% p90 latency and 90% memory reduction.
- **Spearheaded Agent-Tool Gateway**, building a managed connectivity and security layer exposing OpenAPI specs and MCP servers as discoverable agent tools with protocol translation, **80% onboarding reduction per integration.**
- **Designed and developed** a multi-tenant **sandboxed code execution** environment for Code Interpreter and Hosted Shell Tool; authored a **Go-based sandbox supervisor** with **gVisor runtime isolation**, and **mitmproxy-controlled egress**, to safely run untrusted LLM generated code for **Tier-1 enterprise customers**.
- **Engineered** a **warm-pool provisioning strategy** using atomic Kubernetes label updates against a Deployment-managed pod fleet, delivering **sub-100ms container provisioning** on the hot path; built a custom **HPA scaling on lease velocity** instead of CPU to grow capacity ahead of demand and prevent pool exhaustion under traffic spikes.
- **Led a team of 5 engineers** to deliver a provider-agnostic model-selection introducing per-agent and per-tool-path model choice (routing, RAG, SQL generation) where none existed. Abstracted 5 model providers and customer-dedicated GPU clusters behind a unified adapter interface, cutting new-model onboarding from roughly 2 weeks to 1 day.
- **Reduced infrastructure costs by 60%** by consolidating Kubernetes nodes from 500 to 150 through microservice co-location and built a novel phased rollout CRD that progressively deploys new agent versions to canary instances before production, eliminating big-bang deployments and reducing incident risk increasing availability from **95% to 99.9%** .
- **Improved RAG pipeline throughput 4.5x b**y profiling and eliminating bottlenecks using Locust; fixed FastAPI async IO thread starvation, right-sized GPU allocations for embedding and generation LLMs, and replaced synchronous I/O calls with non-blocking patterns, reducing end-to-end latency from **35 seconds to under 15 seconds at 500 QPS.**
- **Built** a **decentralized rate-limiting library** using **gossip protocols over headless Kubernetes services** to synchronize data across pods, eliminating central coordinator bottlenecks and enforcing fair **multi-tenant resource isolation** at scale.

Member of Technical Staff | Oracle Cloud Infrastructure, Seattle, WA ***April 2021 – July 2022***

- Developed a microservice code generation framework that **automated control plane** service scaffolding across all of Oracle Cloud, delivering out-of-the-box application code, infrastructure modules, and observability configuration saving **100+ developer hours** per team for every new OCI Control Plane services.
- **Modernized** legacy cloud workflows using the **Strangler Fig Pattern**, scaling to **5K+ orders per minute** and cutting deployment cycles by **40%** through microservice decomposition.

Software Development Engineering Intern | Amazon, CA ***June 2020 – September 2020***

- **Built** a **debugging console** for on-call engineers to analyze error logs and generate diagnostic reports, reducing data-mismatch investigation time by **10 hours/week**.

Software Developer | SAP, Bangalore, India ***August 2016 – August 2019***

- **Led** the design and development of SAP's **Demand-Driven Material Requirements Planning (DDMRP)** application, deployed on-premise and on **SAP HANA Cloud** and adopted by **Fortune 500 customers** across smart fashion and manufacturing to drive on-time delivery and supply-chain optimization.

PROJECTS

- Implemented **sharding** and a **two-phase-commit**-optimized consensus protocol for the open-source ResilientDB distributed database (later an Apache Incubator project), scaling throughput ~3× to **240K txns/sec**.
- **Developed** a general QA Agent leveraging platform's own agent framework to automate end-to-end regression testing across diverse scenarios and edge cases accelerating release velocity and safeguarding code service quality across GenAI.

EDUCATION

MS Computer Science | University of California Davis | March 2021

BE Information Technology | Pune Institute of Computer Technology | July 2016